

waph-nagpalvv

WAPH-Web Application Programming and Hacking

Lab 0 Report

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Short-bio: I love programming and building stuff. I took this course to learn how to build stuff and add security for the stuff I build, including websites, machine learning models and so much more.



Figure 1: Viplav's headshot

Lab 0 - Development Environment Setup

Overview:

This lab is covered in Lecture 2, with preparation homework in Lecture 1. In Part I, we set up an Ubuntu 22.04 Virtual Machine on VirtualBox and install

software and applications. In Part II, we cloned the course repository and our private repository and complete the `git` exercises to write this report.

The lab's overview:

Lab 0 taught us how to install Ubuntu 22.04 and set it up on the virtual machine. We also learned about libraries like `pandoc` which converts a markdown file to a pdf. We also created a our private repo which we cloned by generating an SSH key first. We also learnt how to clone a public repository by cloning the course repository. We learnt how to write in markdown while preparing this lab report. **Link to this lab report:** [<https://github.com/nagpalvv/waph-nagpalvv/tree/main/labs/lab0>]

Part 1:

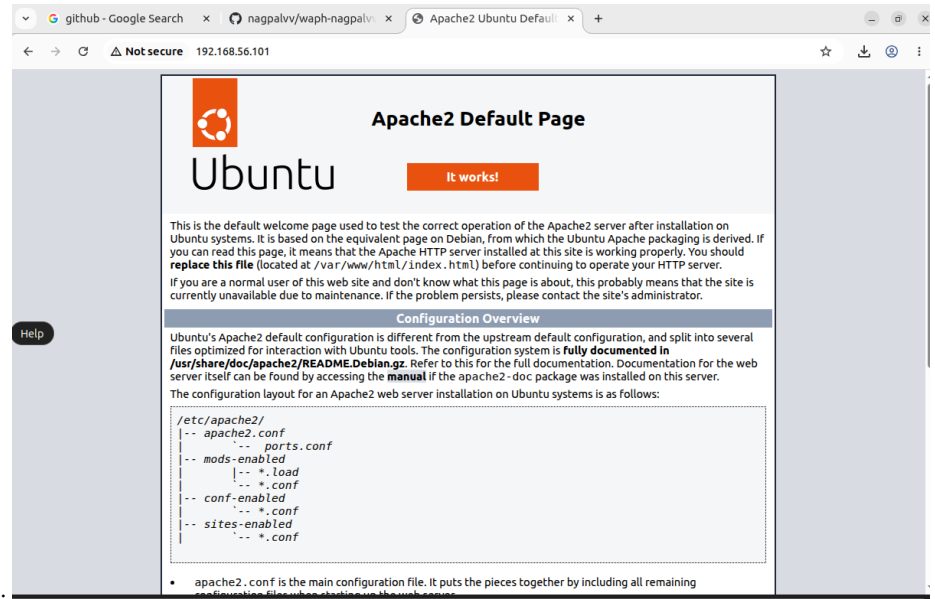
Step 1:

Installed Ubuntu 22.04 on our laptops

Step 2:

Installed VirtualBox by Oracle on our laptops and set up a virtual machine by using the Ubuntu OS ### Step3: After installation, we downloading some pre-requisite libraries like `apache` and `pandoc` and also set up `chrome` to be able to acces the web easier and done!

Apache Web Server Testing:



Apache Test on VM:

Apache Test on Laptop:



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

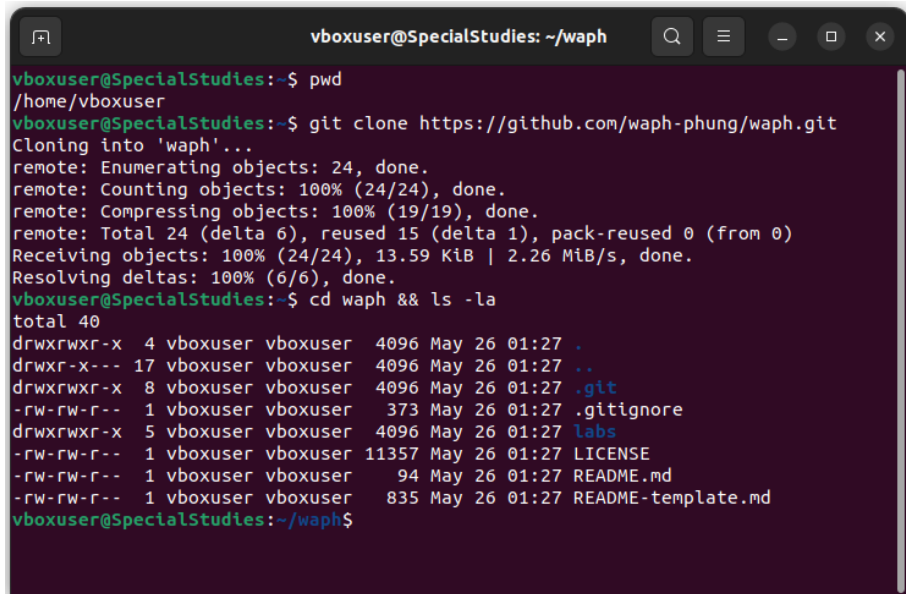
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf`. See their respective man pages for detailed information.
- The binary is called `apache2` and is managed using `systemd`, so to start/stop the service use

Part II - git Repositories and Exercises

The course repository

A terminal window titled 'vboxuser@SpecialStudies: ~/waph' showing the process of cloning a repository and listing its contents. The user runs 'pwd' showing the current directory is '/home/vboxuser'. Then they run 'git clone https://github.com/waph-phung/waph.git'. The terminal shows progress for cloning, including enumerating, counting, and compressing objects. After cloning, they run 'cd waph && ls -la', which displays a directory listing for the 'waph' directory, including files like '.git', '.gitignore', 'labs', 'LICENSE', 'README.md', and 'README-template.md'.

```
vboxuser@SpecialStudies:~$ pwd
/home/vboxuser
vboxuser@SpecialStudies:~$ git clone https://github.com/waph-phung/waph.git
Cloning into 'waph'...
remote: Enumerating objects: 24, done.
remote: Counting objects: 100% (24/24), done.
remote: Compressing objects: 100% (19/19), done.
remote: Total 24 (delta 6), reused 15 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (24/24), 13.59 KiB | 2.26 MiB/s, done.
Resolving deltas: 100% (6/6), done.
vboxuser@SpecialStudies:~$ cd waph && ls -la
total 40
drwxrwxr-x  4 vboxuser vboxuser 4096 May 26 01:27 .
drwxr-x--- 17 vboxuser vboxuser 4096 May 26 01:27 ..
drwxrwxr-x  8 vboxuser vboxuser 4096 May 26 01:27 .git
-rw-rw-r--  1 vboxuser vboxuser  373 May 26 01:27 .gitignore
drwxrwxr-x  5 vboxuser vboxuser 4096 May 26 01:27 labs
-rw-rw-r--  1 vboxuser vboxuser 11357 May 26 01:27 LICENSE
-rw-rw-r--  1 vboxuser vboxuser   94 May 26 01:27 README.md
-rw-rw-r--  1 vboxuser vboxuser  835 May 26 01:27 README-template.md
vboxuser@SpecialStudies:~/waph$
```

Private Repository

Critical steps to creating a private repository:

To create a private repository I navigated to GitHub and created a new repository named waph-nagpalvv ensuring that the visibility was set to **Private**. After creating the repository, I navigated to the repository's **Settings** tab, selected Collaborators, and added the user phung-waph" to grant them access to the repo.

Repository URL: [https://github.com/nagpalvv/waph-nagpalvv.git]

Summary of Hands-on Exercises:

Generating SSH keys: Opened the terminal on the VM and used "ssh-keygen" command to generate a new SSH ey pair.

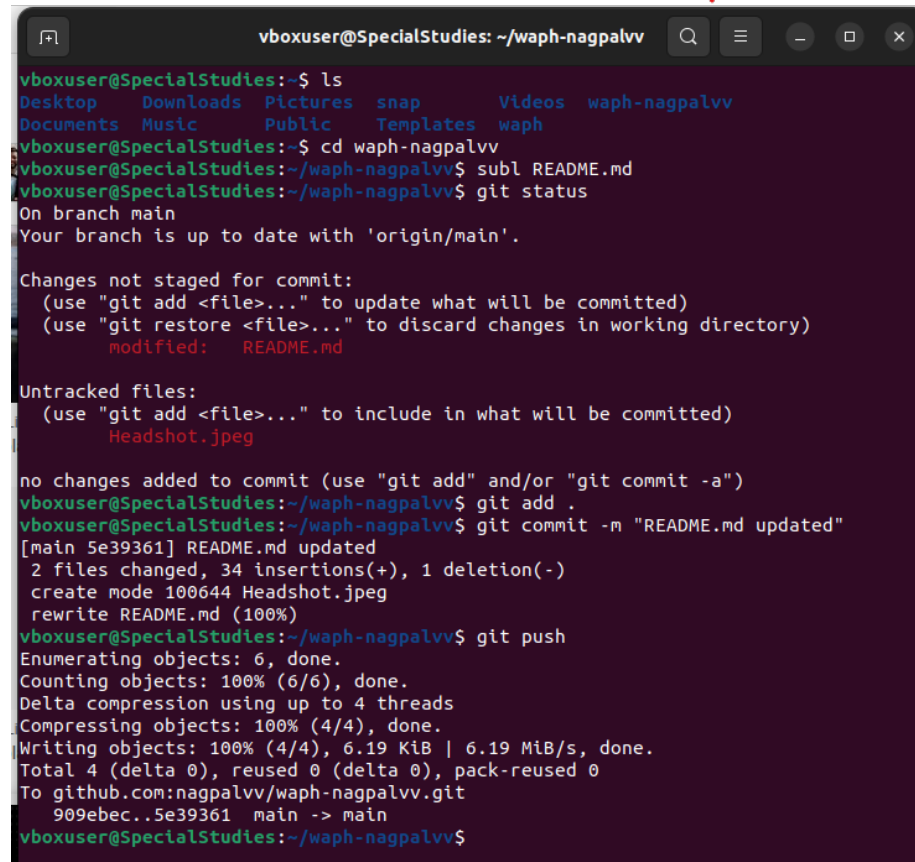
Adding Public key to GitHub: I viewed my public ey using command cat ~/.ssh/id.rsa.pub and copied the output and added it to my Github account under Settings > SSH > New SSH key to establish a secure connectiong between my VM and GitHub.

Cloning Repository: Using the terminal, I ran the git clone command along with my repositorys URL to download a local copy of the remote repository into

my VM.

Editing the README.md file: I opened the README.md file in a text editor and updated the provided template with my personal information.

Committing Changed:

A terminal window titled 'vboxuser@SpecialStudies: ~/waph-nagpalvv' showing a series of git commands and their outputs. The user lists files, changes directory to 'waph-nagpalvv', and checks the status. It shows that 'README.md' is modified and 'Headshot.jpeg' is untracked. The user stages all changes with 'git add .' and commits with 'git commit -m "README.md updated"'. The push command is then executed, showing progress for enumerating, counting, compressing, and writing objects, and finally pushing to the remote repository.

```
vboxuser@SpecialStudies:~$ ls
Desktop  Downloads  Pictures  snap      Videos  waph-nagpalvv
Documents Music      Public   Templates waph
vboxuser@SpecialStudies:~$ cd waph-nagpalvv
vboxuser@SpecialStudies:~/waph-nagpalvv$ subl README.md
vboxuser@SpecialStudies:~/waph-nagpalvv$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Headshot.jpeg

no changes added to commit (use "git add" and/or "git commit -a")
vboxuser@SpecialStudies:~/waph-nagpalvv$ git add .
vboxuser@SpecialStudies:~/waph-nagpalvv$ git commit -m "README.md updated"
[main 5e39361] README.md updated
 2 files changed, 34 insertions(+), 1 deletion(-)
 create mode 100644 Headshot.jpeg
 rewrite README.md (100%)
vboxuser@SpecialStudies:~/waph-nagpalvv$ git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 6.19 KiB | 6.19 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:nagpalvv/waph-nagpalvv.git
 909ebec..5e39361  main -> main
vboxuser@SpecialStudies:~/waph-nagpalvv$
```